

Enterprise Case Studies: A Technical Overview

Rocco DiValentino
Consulting Engineer

3Com
More connected.

Enterprise Case Studies Agenda

PalmSource 99

- Discuss case study selections
- Overview of three customer case studies
- Examine the following:
  - Customer goals
  - Existing business process
  - New business process
  - Technical architecture
  - Architecture components
  - Project outcome
- Q&A

Enterprise Case Studies

PalmSource 99

Selection Criteria

- Demonstrate custom “non-PIM” solutions
- Ability to identify customer and partners
- Include a broad selection of partner technologies
- Obvious differentiation over previous methods
- Different business areas and different industries

Enterprise Case Selections

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- Volvo Cars
  - Automobile Manufacturing
  - Quality inspections and reporting
- United Center Sports Complex in Chicago
  - Entertainment and Retail Sales
  - Inventory management and point-of-sale
- United States Postal Service
  - Mail and package delivery
  - Vendor performance monitoring

Volvo Cars



PalmSource 99

- Sweden-based provider of quality automobiles
- Over 125,000 automobiles enter the United States annually through four different ports
- Each automobile is inspected when unloaded from transport ship
- Each inspection includes up to 40 separate quality checks recorded on paper forms
- Based on quality information from all areas of Volvo, inspection checks are constantly updated to insure highest levels of quality control

Volvo Project Goals

Quality Inspection

- Reduction of paper processing
- Eliminate data entry costs
- Enable query and data analysis
- Even better quality



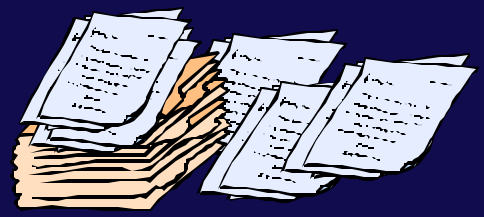
Volvo Inspection—Previous Process

PalmSource 99



Quality inspection at port of entry

- **Paper forms**
- **17 digit chassis number**
- **Inspection of 30–40 items**
- **Risk of damage from clipboards**



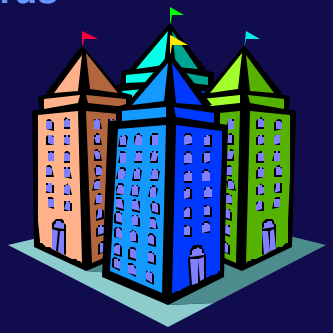
Inspection forms were manually sorted

- **Car Model**
- **Defects**



Out-sourced data entry service

- **Paper forms entered into database**
- **Could take up to three weeks**



Data returned to Volvo headquarters

- **Data could be weeks old**

Volvo Inspection—Paper Forms

PalmSource 99

Clipboards for forms are a potential quality issue

VIN: Y V 1 [ ][ ][ ][ ] X [ ][ ][ ][ ][ ][ ][ ][ ] DATE [ ][ ][ ]



INTERIOR LIGHTING	☐	3540 VISITOR VANITY LIGHT	☐	3541 FRONT READING LIGHTS
	☐	3543 REAR READING LIGHTS		
WARNING LIGHTS	☐	3814 LOW FUEL LEVEL	☐	3948 CHECK ENGINE
	☐	0940 SRS (AIRBAG)	☐	2900 EFI (ELECTR. FUEL INJECTION)
	☐	5930 ABS (ANTI LOCK BRAKES)	☐	3800 CI (COMBINED INSTRUMENT)
	☐	8700 ECC (ELECTR. CLIM. CONTROL)	☐	2750 CC (CRUISE CONTROL)
SOUND SYSTEM	☐	3930 RADIO AM/FM FUNCTION	☐	3931 CASSETTE PLAYER FUNCTION
	☐	3960 SPEAKER DASH CENTER	☐	3961 ANTENNA FUNCTION
CLIMATE CONTROL	☐	8734 COOLING		

Volvo Inspection—New Process

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Quality inspection at port of entry

- **Palm V™ device and electronic forms**
- **Reduced risk of damage**
- **Environmentally friendly rechargeable batteries**



Data sent electronically to Volvo headquarters

- **Inspection data now available in one day**
- **Timely data allows for rapid inspection adjustments**



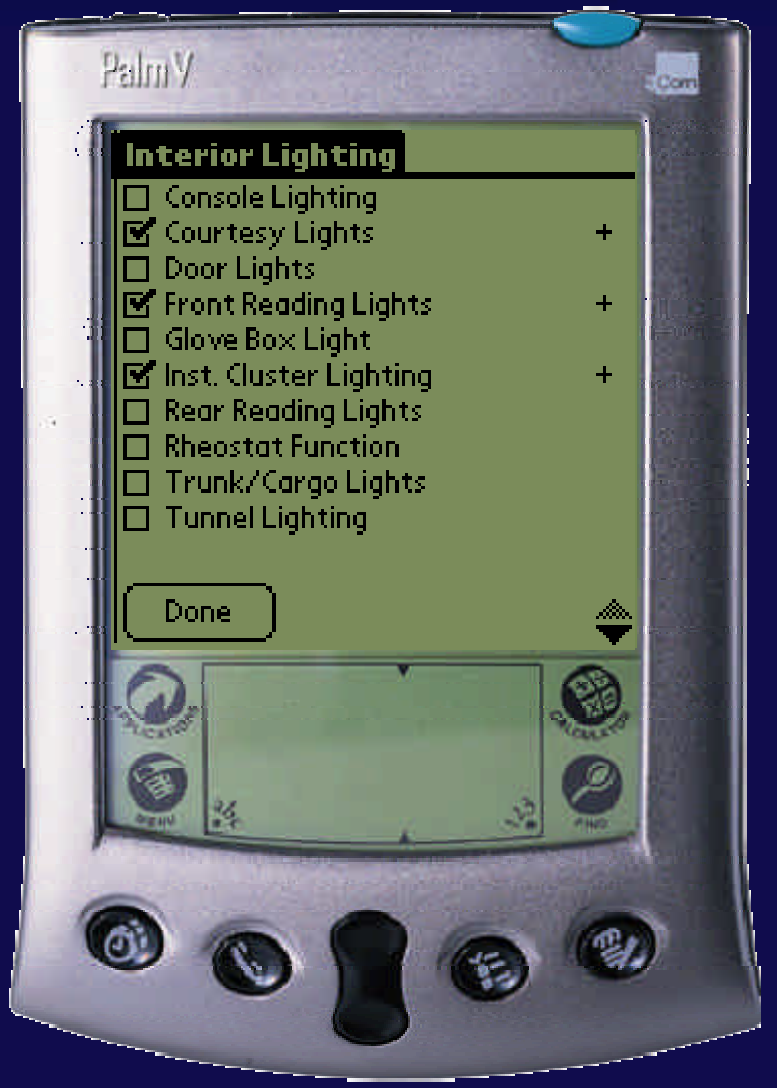
Palm V™ device, HotSync® technology with desktop

- **Data downloaded**
- **New inspections forms loaded**
- **Data files can be immediately analyzed**

Volvo Inspection Application

PalmSource 99

- Metrowerks CodeWarrior C++
- Custom Visual C++ conduit
- Quality data output down to Excel files
- Inspection list uploaded from Excel files
- Users currently synchronize at a single desktop
- Inspection manager sends data directly to Volvo corporate



Volvo Inspection Data

PalmSource 99

Inspection Codes w/description

	A	B	C
1	**Code**	**Description**	**Category**
2	3521	Low Beam Left	Exterior Lighting Front
3	3522	Low Beam Right	Exterior Lighting Front
4	3524	High Beam Left	Exterior Lighting Front
5	3622	High Beam Right	Exterior Lighting Front
6	3561	Fog Light Left	Exterior Lighting Rear
7	3562	Fog Light Right	Exterior Lighting Rear
8	3631	Rear Light Left	Exterior Lighting Rear
9	3632	Rear Light Right	Exterior Lighting Rear

Checklist by Model

	A	B
1	**Prefix**	**Code**
2	YV1	8410
3	YV1	8411
4	YV1	8412
5	YV1	8413
6	YV1TS	3524
7	YV1TS	3522
8	YV1TS	3539

Quality data from inspections

	A	B	C
1	**VIN**	**Code**	
2	YV1TS5566887799123	3842	
3	YV1TS5566887799123	3952	
4	YV1TS5566887799123	3841	
5	YV1TS5566887799123	5840	
6	YV1TS5566887799123	9999	Right Hand Drive
7	YV1TS5566887799123	comment	Purple with Blue PinStripe

TikiSoft—
Volvo Inspections Developer

PalmSource 99

- Palm Computing® Platform Consulting and development
- Project Manager, Mark Weiss
- Past project include:
  - Volvo
  - American Red Cross
  - Qualcomm
  - U.C. Irvine Department of Psychology



www.tikisoft.com

Volvo Inspection Results

PalmSource 99

- Inspection results immediately available
  - From 3 weeks to same day availability
- New inspection updates easily implemented
- Immediacy of inspection data improves quality through quick notification of quality issues to manufacturing
- Elimination of data entry and shipping costs
- Improved data integrity

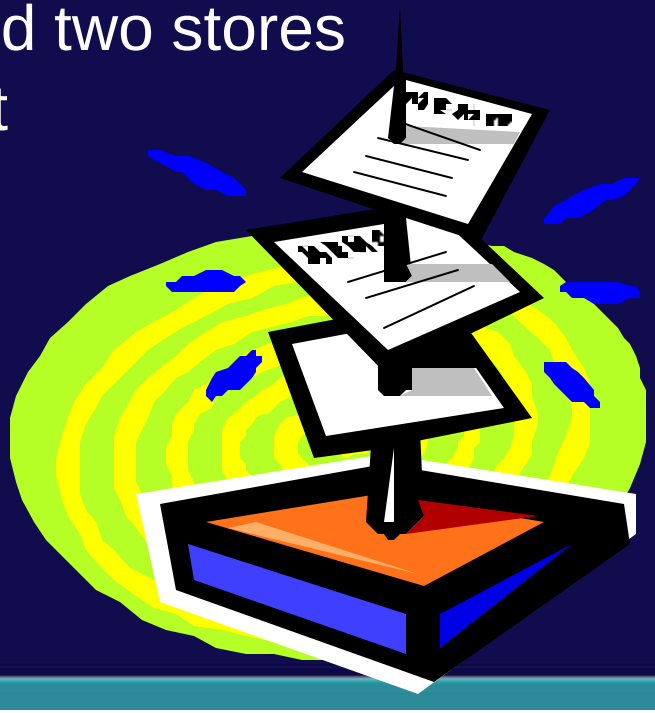
United Center



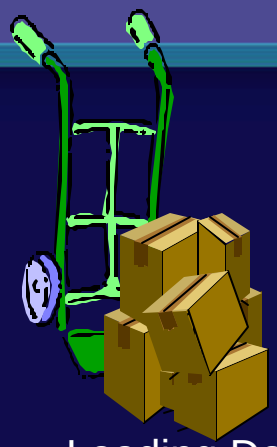
- Home of the Chicago Bulls and Blackhawks
- Retail sales of event merchandise through twelve merchandise stands and two retail stores
- 50,000 to 300,00 items are sold at a single event
- Wide variety of merchandise with no detailed tracking of sales
- All tracking of sales and inventory were paper based and prone to error and lack of control

United Center Goals

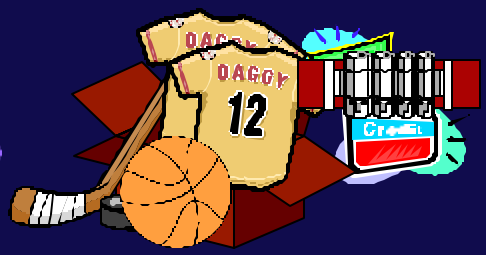
- Provide inventory audit control
  - Reduce on-hand inventory
  - Control inventory loss
- Improve Point-of-Sale (POS) data acquisition
  - Twelve merchandise stands and two stores
  - Better target retail sales market
- Improve cost accounting and inventory reconciliation



United Center—Previous Process







Loading Dock and Warehouse

- **Minimal tracking of lost inventory**
- **Over 10,000 SKUs**

Merchandise Sales

- **“Out of apron” sales**
- **Cash or Credit**
- **No detailed tracking**



Back-end System all paper based

- **Accurate forecasting difficult**
- **Inventory and accounting reconciliation challenging**

United Center—New Process



Loading Dock and Warehouse with mobile inventory app & scanning

- **Incoming goods are scanned**
- **Orders reconciled against receiving**
- **Pick lists filled with bar-coding**
- **Much faster inventory reconciliation**
- **Signature capture and ID badge scan when receiving merchandise**

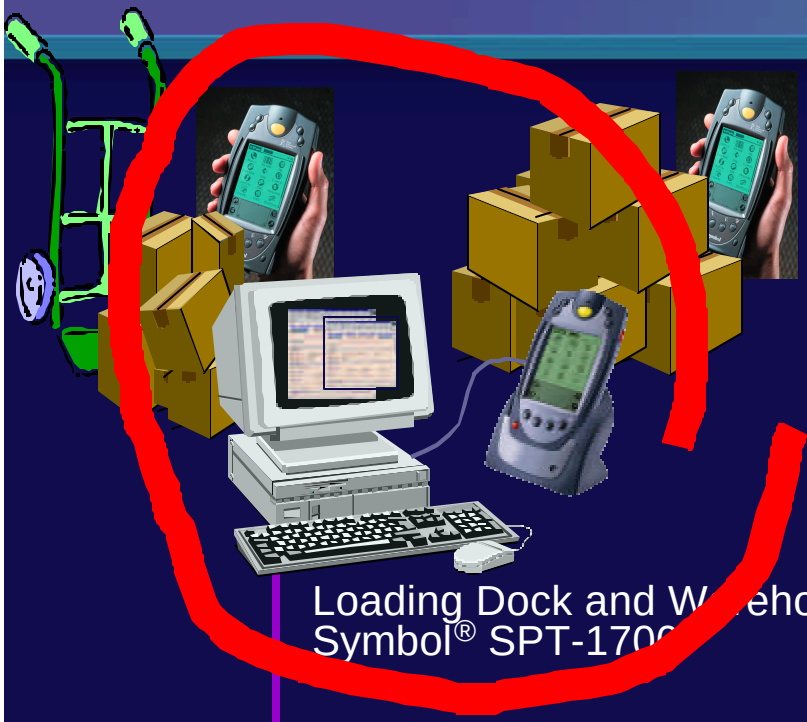
Merchandise Sales with mobile point-of-sale app & scanning

- **All Sales tracked via bar-coding**
- **Accurate SKU and sales tracking**
- **Transfers between stands are tracked**
- **Receipt printing**

Automated back-end Systems

- **Elimination of manual data entry**
- **Detailed sales information available**

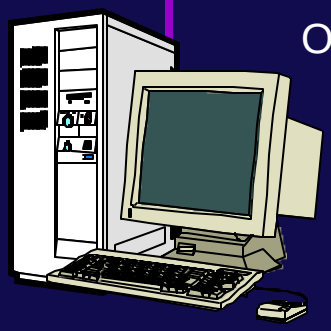
United Center Architecture



Loading Dock and W ehouse
Symbol® SPT-170

Merchandise Sales
Symbol® SPT-1700





Oracle 8 Enterprise Server



Intranet Workstations

Symbol® SPT-1700 Hardware

PalmSource 99

- Palm Computing® platform
- Laser Barcode scanning
- Ruggedized
- Weather resistant
- LiION Battery Packs
- 2-8MB RAM, 2–4MB ROM
- Charging cradle



Palm OS® Inventory Application

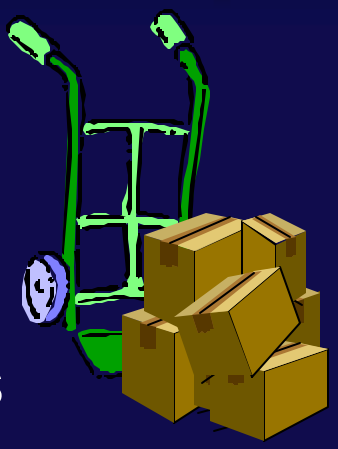
- Puma Satellite Forms™ 3.1
  - Symbol scanning extension
  - Oracle compatible tables
- Oracle® 8iLite Desktop
- Oracle® 8iLite Consolidator
  - Programmable conduit
  - Synchronize Satellite Forms™ tables with 8iLite



Oracle8i

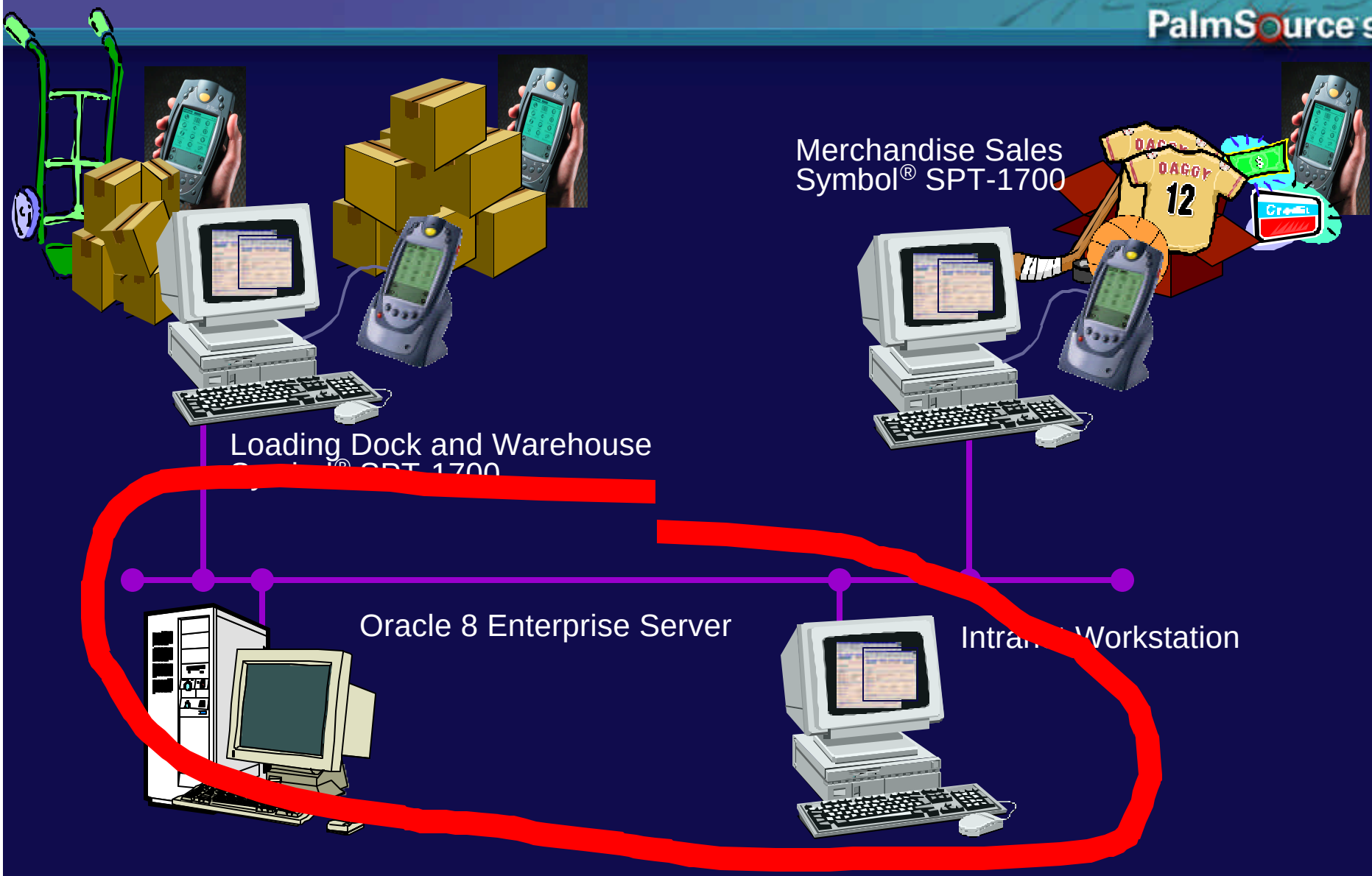
Satellite Forms Enterprise Edition

Inventory Application Functions



- Inventory reconciliation
- Receiving
  - Synchronize purchase order from Oracle
  - Reconcile purchase order with Shipments
  - Scan all incoming items
- Issuance
  - Sync “pick list” from stands and re-supply retail outlets
  - Inventory reconciliation (warehouse, stand, and store)
  - Stand to stand transfers supported
  - Badge scan and signature capture at delivery

Intranet Server and Workstation



United Center Server

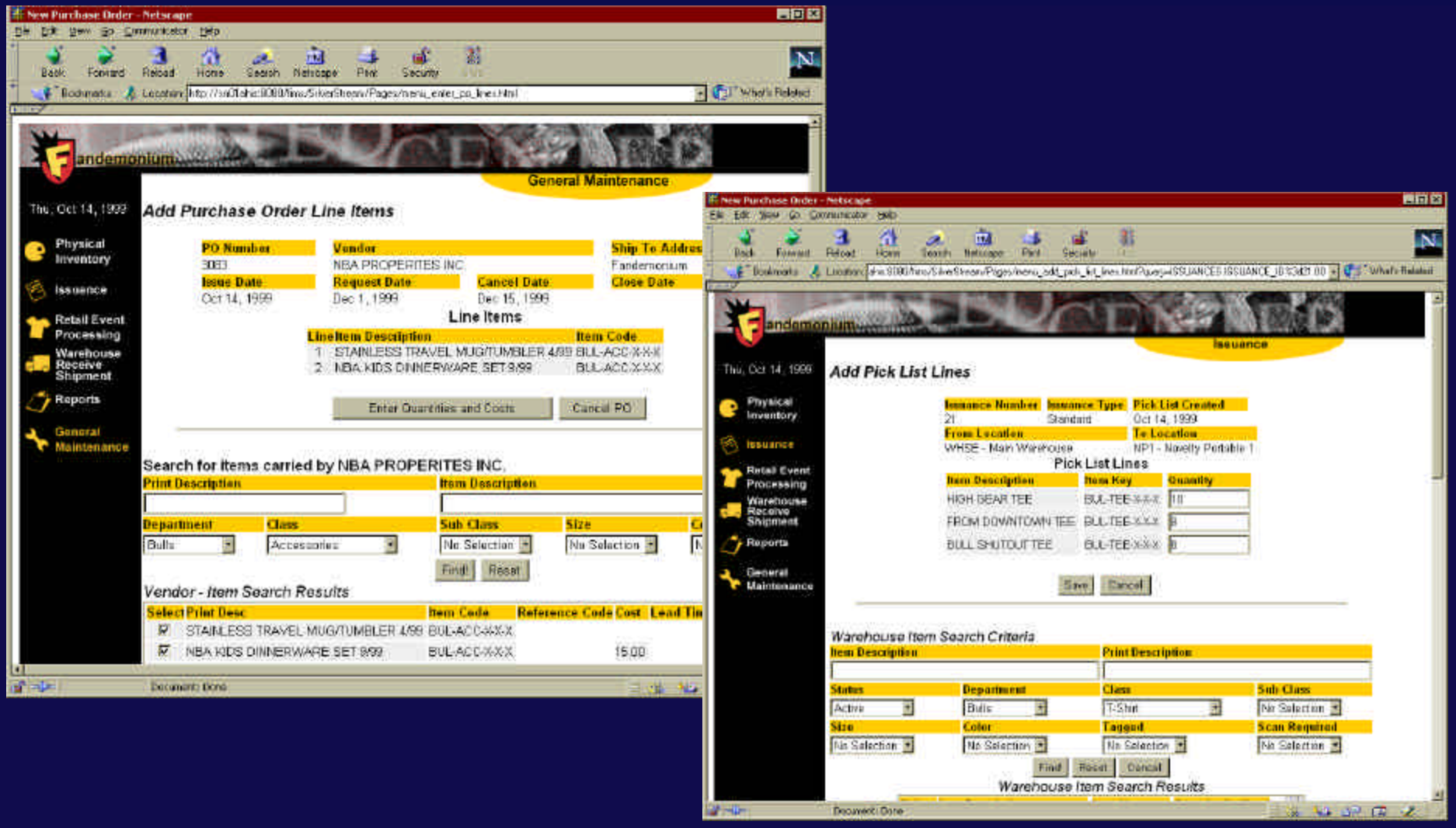




- Silicon Graphics intranet server
- Oracle 8 Enterprise
  - Replicates with 8iLite on Desktops
- SilverStream (Web server rapid application development tools)
  - Java
- Platinum Financials
  - Custom inventory management
  - Custom purchasing module

United Center Intranet Apps

PalmSource 99



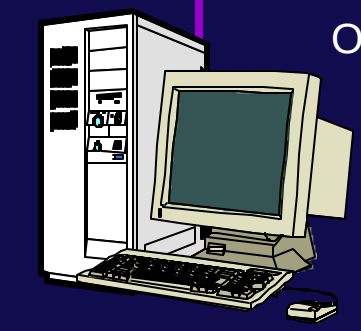
Point of Sale Application



Loading Dock and Warehouse
Symbol® SPT-1700



Merchandise Sales
Symbol® SPT-1700



Oracle 8 Enterprise Server



Intranet Workstation

Palm OS® Point of Sale Application

PalmSource 99

- Puma Satellite Forms™ 3.1
  - Symbol scanning extension
  - Bachmann printing extension
    - Comtec Encore 2 2” printers
    - Serial connection (future I/R)
  - Oracle compatible tables
- Oracle® 8iLite Desktop
- Oracle® 8iLite Consolidator
  - Programmable conduit
  - Synchronize Satellite Forms™ tables with 8iLite



Oracle8i

Satellite Forms Enterprise Edition

Point of Sale Functionality

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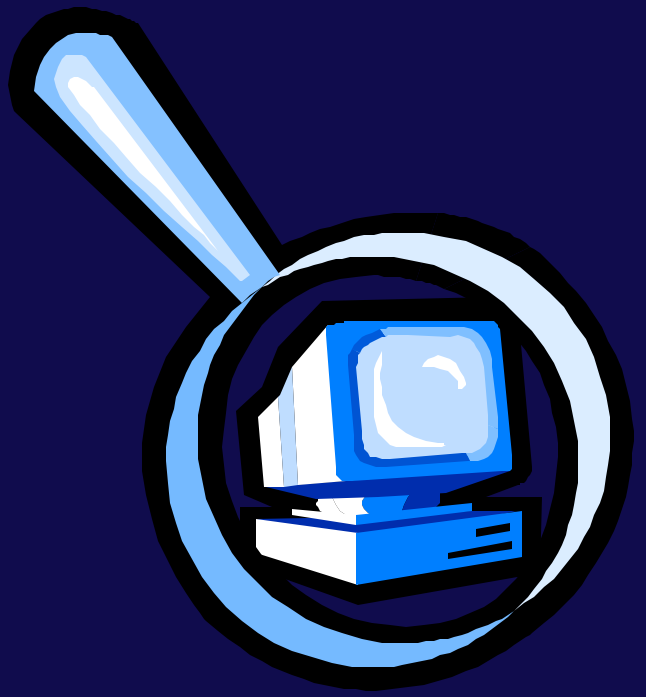
- Sales
  - Bar-coding provides accurate sales data
  - Sales data used to generate accurate “pick lists”
  - Support for printer receipts
- Issuance
  - Stand to stand inventory transfers
- Close Out
  - No total cash or credit available to user
  - User enters cash/credit on hand
  - Unit is locked unit synchronized at cashier
  - Cashier takes cash/credit and checks for balance



United Center Development Specifics

PalmSource 99

- Developer's first Palm OS® software project
  - One Palm OS developer
  - One SilverStream developer
  - One database designer
  - One Oracle reports developer
- Project start Aug. 10 '99
- Project completion Oct. 8 '99
  - Three weeks requirements
  - Two weeks design
  - Nine weeks total



Braxton Butterfield Consulting, Inc.

United Center Project Management & Development

- Application architecture and development
- Web development
- Database and System administration
- Headquarters in Arlington Heights, IL
- Project Manager, John Simon

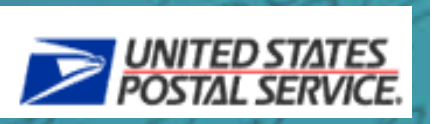
www.bbcg.com

United Center Fandemonium Results

- Reduction in on-hand inventory
- Inventory reconciliation dramatically reduced
  - Dropped from 48–72 hrs down to 4–6 hours
- Complete accountability of merchandise
- Able to better predict and replenish merchandise
- Improved customer satisfaction



United States Postal Service



PalmSource 99

- USPS delivers hundreds of millions of letters and packages worldwide
- Commercial airlines are contracted to provide mail transport for over $1 billion per year
- Ramp clerks monitor airline performance and create incident reports when problems occur
- The data from incident reports is used to manage routes, improve performance, and assess fines
- The existing paper based system did not provide timely reporting compromising service and collection of fines

United States Postal Service Goals

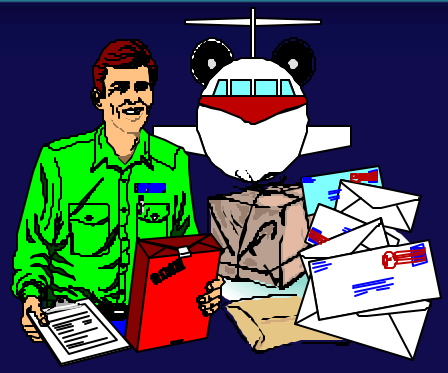
PalmSource 99

Irregular Mail Tracking System (IMTS)

- Improve/simplify data capture
- Improve accuracy of information
- Improve timeliness of information
- Improve reports and access to details
- Reduce maintenance and deployment costs
- Ultimately improve carrier performance

USPS—
Previous Process and Methods

PalmSource 99



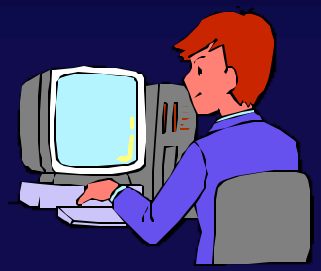
Data collection using paper forms.

- Time consuming
- Prone to errors



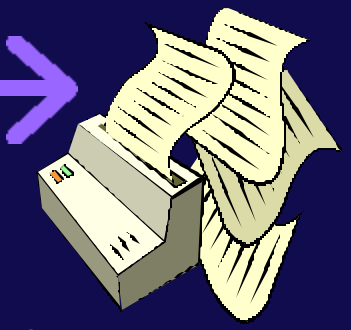
Forms pile up for days and weeks

- Delays process



Data entry from Forms

- Time consuming
- Prone to errors
- Summary info only
- Local database only!



Paper forms & reports generated

- Still some hand written forms

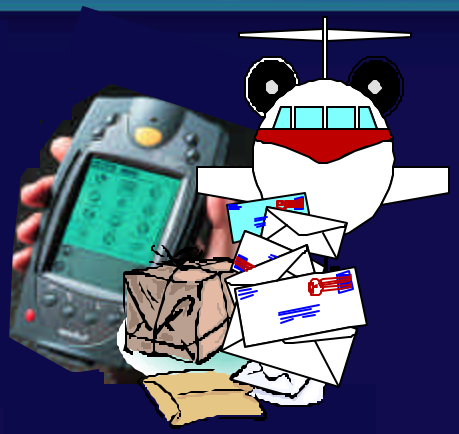


Adjudication with carrier

- Up to 3 months after the fact
- Timely remediation difficult

USPS—
New Process and Methods

PalmSource 99



Data collection with scanning and electronic forms

- More efficient
- Reduced errors

Future ExtraNet



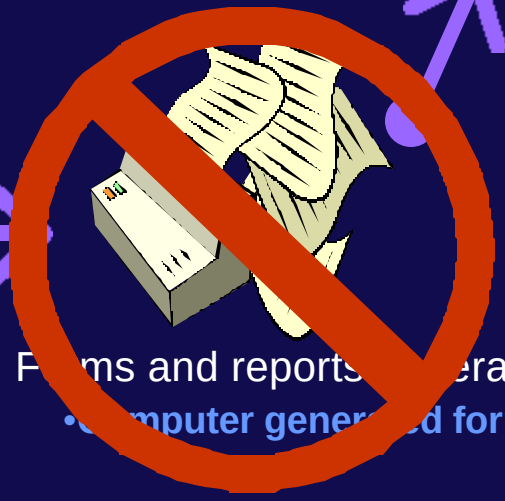
Adjudication with carrier

- Timely
- Scanned data is more accurate
- Faster remediation



Direct synchronization with central database

- Immediate access via USPS Intranet
- Access to details and OLAP
- Reduces data entry



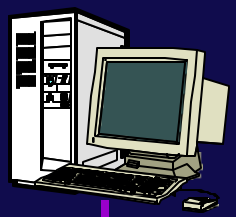
Forms and reports generated

- Computer generated forms

USPS—
Irregular Mail Tracking System

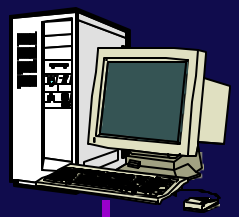
PalmSource 99

Synchronization & Web Server



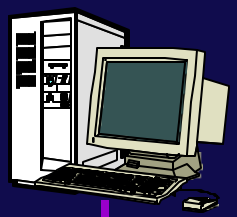
- ScoutSync™
- MS IIS™ 4.0
- Intranet (ASP)
- Microsoft® MTS

Cognos® Web & OLAP Server



- OLAP Cubes
- MS IIS™ 4.0

Database Server



- SQL Server 7.0
- ODBC
- Moving to Oracle®

USPS Headquarters — Washington, DC

Postal Service Network

Airport Mail Center — Area Postal Center

Workstation



- IE 4.X
- Access to IMTS
- Access to Cognos®

Symbol® -1700s

- 4 bay Ethercradle



Workstation



- IE 4.X
- Access to IMTS
- Access to Cognos®

IMTS Mobile Data Collection

PalmSource 99

Synchronization & Web Server

- ScoutSync™
- MS IIS™ 4.0
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Workstation

- IE 4.X
- Access to IMTS
- Access to Cognos®

Symbol® SPT-1700 Hardware

- Palm Computing® platform
- Laser barcode scanning
- Ruggedized
- Weather resistant
- LiION Battery Packs
- 2–8MB RAM, 2–4MB ROM
- 4 bay Ethercradle
- Spectrum 24 optional
  - Wireless LAN
  - Real-time data collection



IMTS Incident Capture Application

PalmSource 99

- MetroWerks ® C++
  - Symbol® SDK
  - Palm OS® data manager
- Palm OS® emulator
  - Development testing
  - End user testing
- Intuitive interface
- ScoutSync™ client
  - Included in ROM
- “Auto-Sync” in cradle



IMTS Synchronization and Web Server

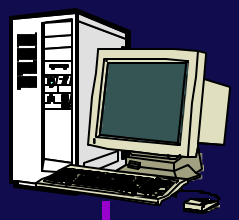
PalmSource 99

ynchronization & Web Serve



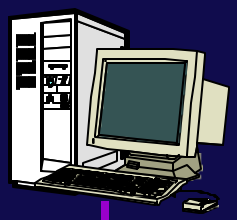
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Symbol® -1700s

- •4 bay Ethercradle



Workstation



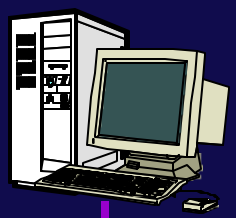
- • IE 4.X
- • Access to IMTS
- • Access to Cognos®

IMTS Synchronization Server



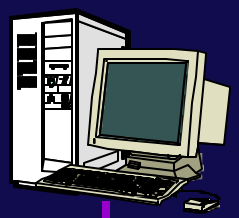
- Riverbed Technologies' ScoutSync™
- Microsoft IIS™ 4.X Intranet Server
- Automatic configuration
  - Default group installs applications during initial sync
  - Application then queries the user for site ID
  - Conduit configures new units with airport specific info
- Automatic distribution of application updates
- Login IDs for each site
  - Used to track possession of units
- Server Conduits developed with Visual C++
  - ODBC interface to SQL Server 7

IMTS Database Server



Synchronization & Web Server

- ScoutSync™
- MS IIS™ 4.0
- Intranet (ASP)
- Microsoft® MTS



Cognos® Web & OLAP Server

- OLAP Cubes
- MS IIS™ 4.0



Database Server

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- 4 bay Ethercradle



Workstation

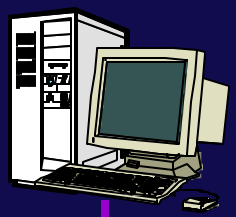


- IE 4.X
- Access to IMTS
- Access to Cognos®

IMTS OLAP Server

PalmSource 99

Synchronization & Web Server



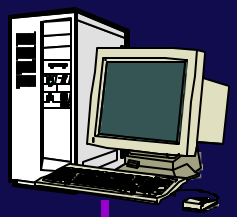
- •ScoutSync™
- •MS IIS™ 4.0
- •Intranet (ASP)
- •Microsoft® MTS

Cognos® Web & OLAP Server



- •OLAP Cubes
- •MS IIS™ 4.0

Database Server



- • SQL Server 7.0
- • ODBC
- • Moving to Oracle®

USPS Headquarters — Washington, DC

Postal Service Network

Airport Mail Center

Area Postal Center

Workstation



- • IE 4.X
- • Access to IMTS
- • Access to Cognos®

Symbol® -1700s

- •4 bay Ethercradle



Workstation

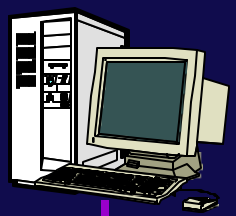


- • IE 4.X
- • Access to IMTS
- • Access to Cognos®

IMTS Workstations

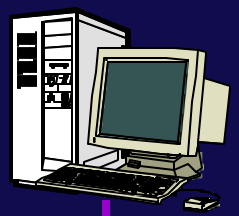
PalmSource 99

Synchronization & Web Server



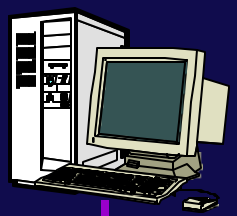
- ScoutSync™
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Cognos® Web & OLAP Server



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Database Server



- SQL Server 7.0
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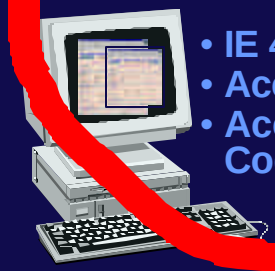
Washington, DC

Postal Service Network

Airport Mail Center

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Workstation



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Workstation

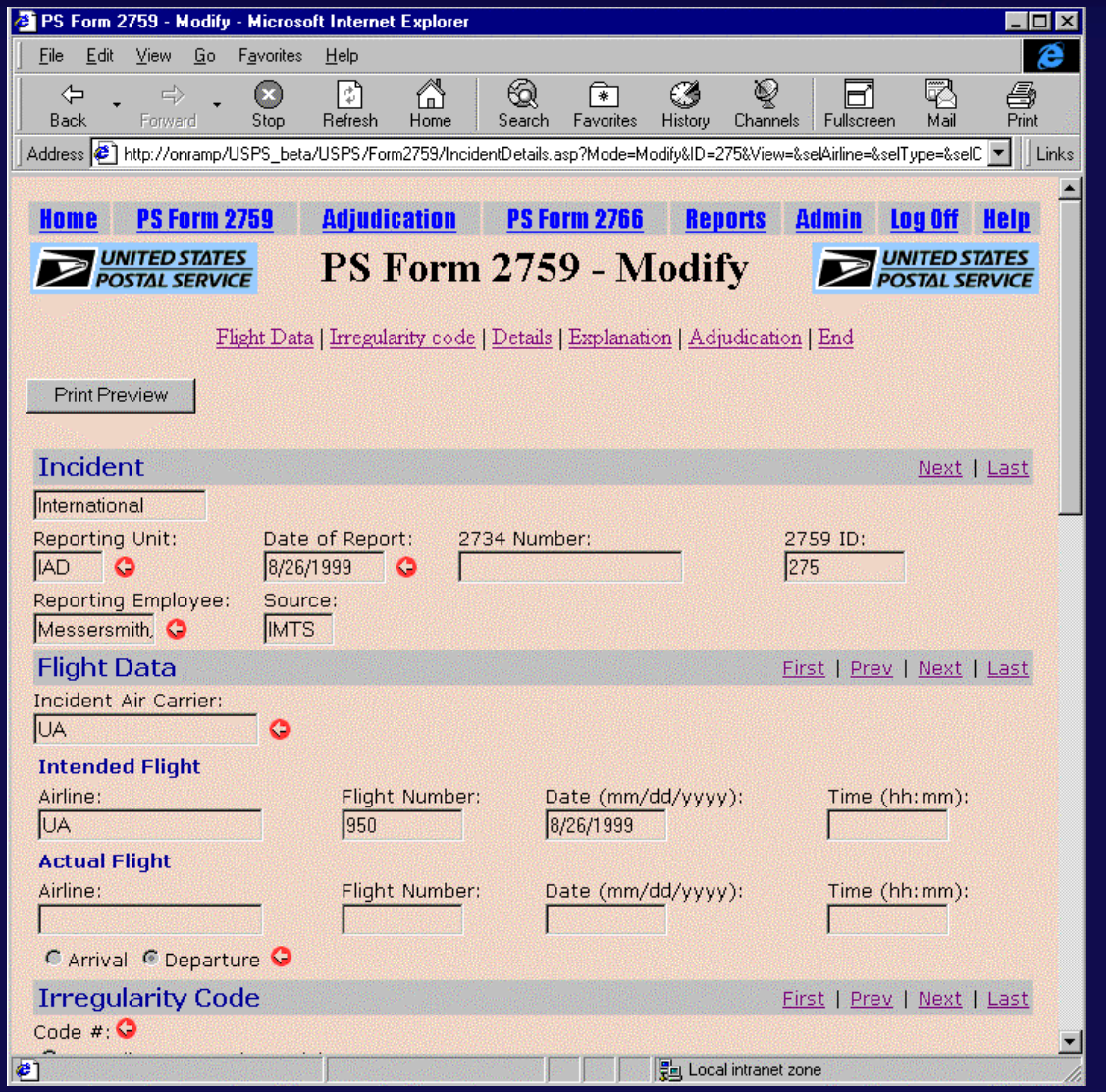


- IE 4.X
- Access to IMTS
- Access to Cognos®

IMTS Intranet Applications

PalmSource 99

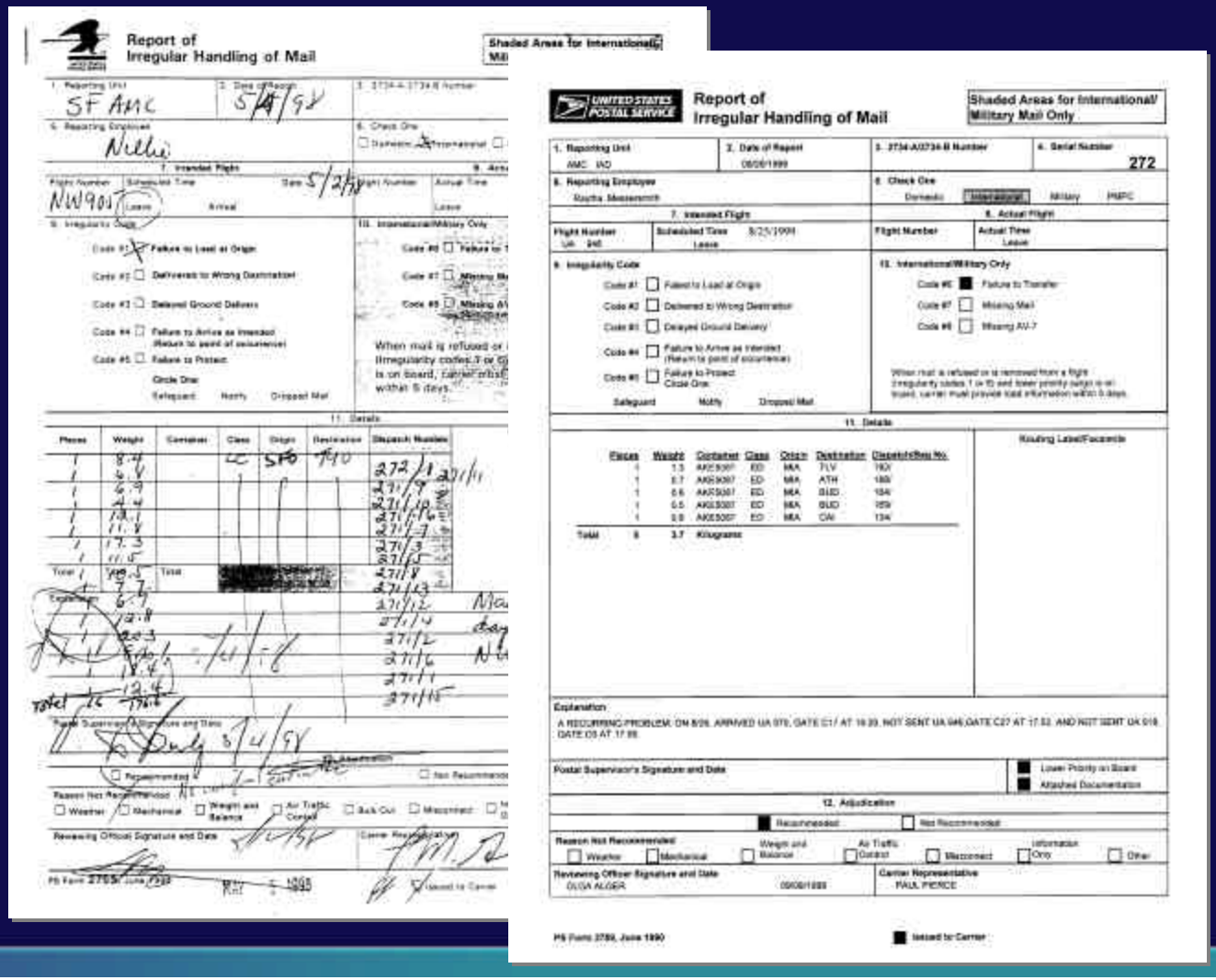
- Administration
- Create Incidents
- Modify Incidents
- Generate reports



IMTS Forms and Reporting

PalmSource 99

Form 1099, Improper Handling of Mail



Report of
Irregular Handling of Mail

Shaded Areas for International/Military Mail Only

1. Reporting Unit: SF AMC
2. Date of Report: 5/4/98
3. 2734-A/2734-B Number:
4. Reporting Employee: Nellie
5. Check One: Domestic / International
6. Intended Flight
7. Actual Flight
8. Irregularity Code
   - Code #1: Failure to Load at Origin
   - Code #2: Delivered to Wrong Destination
   - Code #3: Delayed Ground Delivery
   - Code #4: Failure to Arrive as Intended (Return to point of occurrence)
   - Code #5: Failure to Protect
   - Circle One: Safeguard / Notify / Dropped Mail
9. Flight Number: NW901
   Scheduled Time: Leave / Arrival
   Date: 5/2/98
   Actual Time: Leave
10. International/Military Only
    - Code #6: Failure to Transfer
    - Code #7: Missing Mail
    - Code #8: Missing AV-7
    - When mail is refused or is removed from a flight (irregularity codes 1 or 5) and lower priority cargo is on board, carrier must provide total information within 5 days.
11. Details

Pieces	Weight	Container	Class	Origin	Destination	Dispatch/Rec No.
1	8.4		LC	SFO	TUS	272/1
1	6.4					271/9
1	6.9					271/10
1	4.4					271/11
1	13.1					271/2
1	11.8					271/7
1	17.3					271/3
1	11.5					271/5
Total 1	89.3	Total				271/8
Explanation	7.7					271/13
	12.8					271/12
	20.3					271/14
	5.6					271/2
	18.4					271/6
Total 16	13.4					271/1
	194.6					271/15

Postal Supervisor's Signature and Date: Duly 5/4/98

☐ Recommended ☐ Not Recommended

Reason Not Recommended: ☐ Weather ☐ Mechanical ☐ Weight and Balance ☐ Air Traffic Control ☐ Back Out ☐ Misconnect

Reviewing Official Signature and Date

Carrier Representative

PS Form 3755, June 1990 — Issued to Carrier

UNITED STATES POSTAL SERVICE

Report of
Irregular Handling of Mail

Shaded Areas for International/Military Mail Only

1. Reporting Unit: AMC IAD
2. Date of Report: 05/06/1998
3. 2734-A/2734-B Number:
4. Serial Number: 272
5. Reporting Employee: Sasha Messenmith
6. Check One: Domestic / International / Military / PMPC
7. Intended Flight
8. Actual Flight
   - Flight Number: UA 946
   - Scheduled Time: Leave 5/25/1998
   - Flight Number:
   - Actual Time: Leave
9. Irregularity Code
   - Code #1: Failed to Load at Origin
   - Code #2: Delivered to Wrong Destination
   - Code #3: Delayed Ground Delivery
   - Code #4: Failure to Arrive as Intended (Return to point of occurrence)
   - Code #5: Failure to Protect
   - Circle One: Safeguard / Notify / Dropped Mail
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    - Code #8: Missing AV-7
    - When mail is refused or is removed from a flight (irregularity codes 1 or 5) and lower priority cargo is on board, carrier must provide total information within 5 days.
11. Details

Pieces	Weight	Container	Class	Origin	Destination	Dispatch/Rec No.
1	1.3	AKE5097	ED	MIA	TLV	160
1	0.7	AKE5097	ED	MIA	ATH	188
1	0.6	AKE5097	ED	MIA	BUD	134
1	0.5	AKE5097	ED	MIA	BUD	159
1	0.9	AKE5097	ED	MIA	CAI	134
Total 5	3.7	Kilograms				

Routing Label/Facsimile

Explanation: A RECURRING PROBLEM. ON 8/26, ARRIVED UA 976, DATE C17 AT 16:20. NOT SENT UA 946 DATE C27 AT 17:52. AND NOT SENT UA 918 DATE C5 AT 17:55.

Postal Supervisor's Signature and Date

☐ Lower Priority on Board ☐ Attached Documentation

12. Adjudication

☐ Recommended ☐ Not Recommended

Reason Not Recommended: ☐ Weather ☐ Mechanical ☐ Weight and Balance ☐ Air Traffic Control ☐ Misconnect ☐ Information Only ☐ Other

Reviewing Officer Signature and Date: OLGA ALGER 05/06/1998

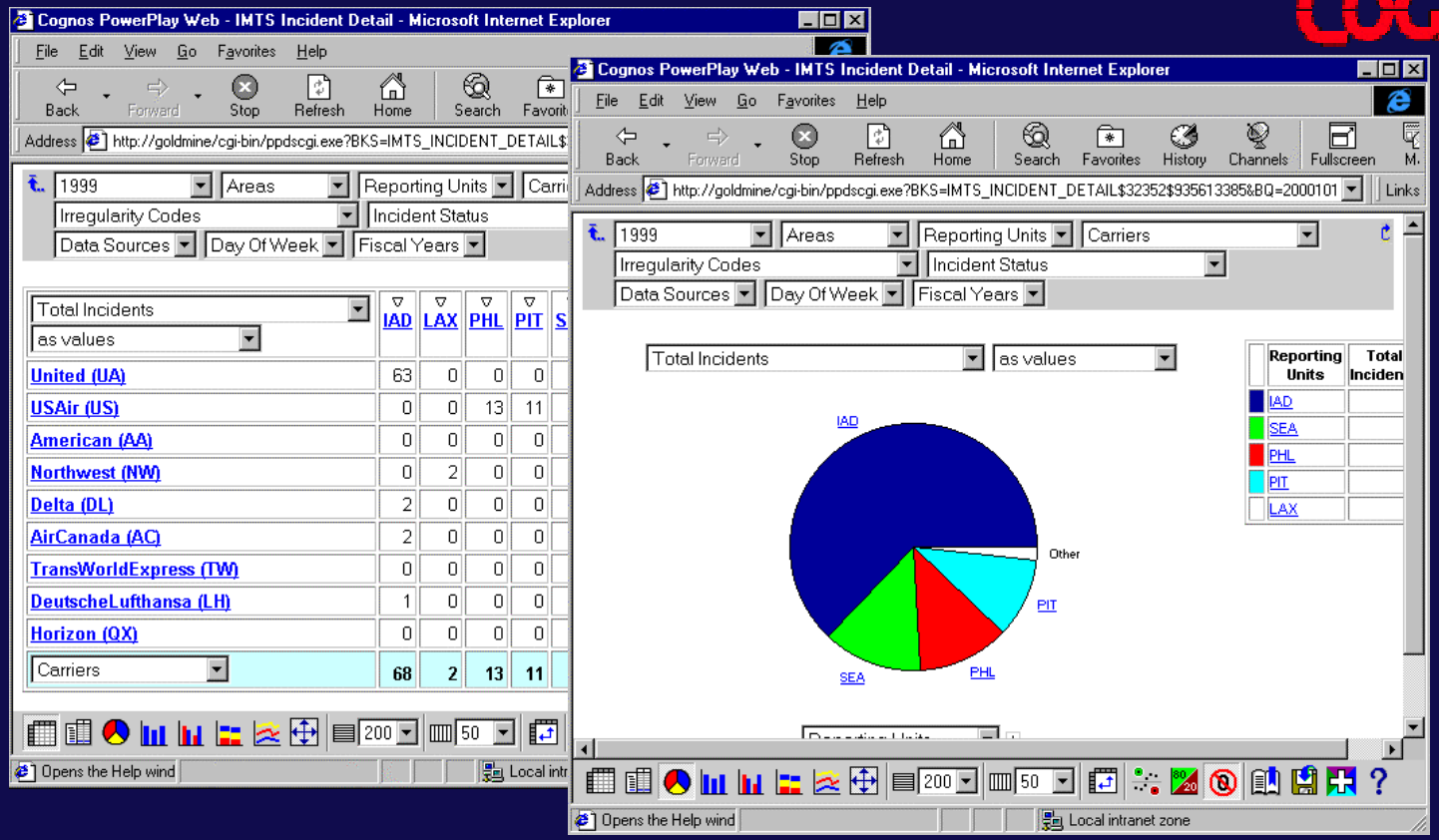
Carrier Representative: PAUL PIERCE

PS Form 3755, June 1990 — ☐ Issued to Carrier

IMTS Analytical Capabilities

PalmSource 99

- Cognos multidimensional data arrays, “Cubes”, allow for system wide incident analysis



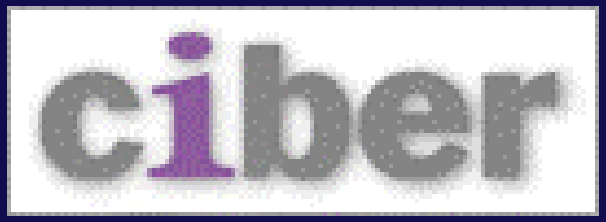
IMTS Development Specifics

- Three member development team
  - Project lead and SQL developer
  - Web developer
  - Palm OS / Server Conduit developer
- Proof of concept
  - 11/98 and12/98, 2 months total
- Project
  - Kick-off January ’99
  - Completed July ’99
- Future phases also planned



Ciber—US Postal Service Developer

- Management and IT Consulting
- 6,700 employees, 47 offices
- USPS project managed from McLean, VA office
- Project Manager, Kevin Norris



www.ciber.com

US Postal Service Results



PalmSource 99

- IMTS solution expanded to 80 airports for domestic, in addition to, international mail
- USPS expects ROI for the 1,000 Symbol® devices within six months
- Future plans to expand use to the airlines

"The ability to easily synchronize our data, from a wide number of locations, is a huge advantage. Now, accurate tracking information can be accessed at any time by managers around the country."

—Clayton Bonnell, USPS Manager of International Operations

Enterprise Case Studies Summary

PalmSource 99

- Reviewed three different case studies
- Each with a different level of complexity
- Discussed technical aspects and business improvements
- Use these as a starting point to consider what the Palm Computing® platform can deliver to your project, improving the way you do business


PalmSource™ 99
Q&A
3Com
More connected.


PalmSource™ 99
3Com
More connected.